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Low profile rotary damper

Abstract

Described is a low profile rotary damper assembly having a rotor, an annular seal, a housing, a bearing plate, and a gear. The rotor includes a shaft portion integrally formed with a body portion. The shaft portion defines an engagement interface at a distal end thereof. The annular seal is configured to surround at least a portion of the shaft portion. The housing defines an opening and a rotor cavity, wherein the housing is configured to rotatably support the rotor. The bearing plate is coupled to the housing and is configured to seal the opening. The bearing plate includes an annular shoulder defining an aperture and a convex cap defining a seal cavity for the annular seal between the body portion and an interior surface of the convex cap. The shaft portion extends axially outwardly through the aperture. The gear is coupled to the rotor via an opening that is configured to receive the engagement interface. The gear is shaped with a generally concave cavity that is sized and shaped to receive the collar and at least a portion of the convex cap.

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Background/Summary

RELATED APPLICATION (1) The present application claims priority to U.S. Provisional Patent Application No. 63/468,160, filed May 22, 2023, and entitled “Low Profile Rotary Damper,” which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Movement dampers of various types are used in assemblies and applications to control the

movement of assembly components. For example, movement dampers can be used to control movement in at least one direction of a drawer or door in furniture, cabinets and appliances. In the automotive field, movement damper can be used on, for example, glove box and counsel doors as well as other doors and closures for bins, storage areas, and the like.

(2) Example movement dampers include rotary dampers. A rotary damper generally comprises a housing and a rotor rotatable in the housing, with a shaft of the rotor extending outwardly from the housing to receive a gear thereon to, in turn, engage a gear on the device being controlled. In some rotary dampers of this type, a viscous fluid is provided in the housing, which is then sealed. Damping performance is facilitated by resistance of the fluid to rotation of the rotor within the sealed housing.

(3) While effective, existing rotary damper designs result in a large profile (e.g., height) and, therefore, are not suitable for applications that impose physical size constraints on the rotary damper. Therefore, despite advancements to date, it would be highly desirable to have a low profile rotary damper that can be used in applications where space is limited.

SUMMARY

(4) The present disclosure relates generally to a movement dampers and, more particularly to rotary dampers substantially as illustrated by and described in connection with at least one of the figures, as set forth more completely in the claims.

Description

DRAWINGS

(1) The foregoing and other objects, features, and advantages of the devices, systems, and methods described herein will be apparent from the following description of particular examples thereof, as illustrated in the accompanying figures; where like or similar reference numbers refer to like or similar structures. The figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the devices, systems, and methods described herein.

(2) FIG. 1a illustrates a topside isometric view of the rotary damper assembly in accordance with an aspect of this disclosure, while FIG. 1b illustrates an underside isometric view thereof.

(3) FIGS. 1c through 1f illustrate, respectively, first, second, third, and fourth elevation side views of the rotary damper assembly.

(4) FIG. 1g illustrates a top plan view of the rotary damper assembly, while FIG. 1h illustrates a bottom plan view thereof.

(5) FIG. 1i illustrates a cross-sectional elevation side view of the rotary damper assembly taken along cutline A-A.

(6) FIG. 1j illustrates a cross-sectional isometric view of the rotary damper assembly taken along cutline A-A.

(7) FIG. 1k illustrates an elevation side assembly view of the rotary damper assembly, while FIG. 1l illustrates a cross-sectional view thereof taken along cutline A-A.

(8) FIG. 1m illustrates an isometric assembly view of the rotary damper assembly, while FIG. 1n illustrates a cross-sectional view thereof taken along cutline A-A.

(9) FIG. 2a illustrates a bottom plan view of the gear of FIG. 1a.

(10) FIG. 2b illustrates a top plan view of the rotor of FIG. 1a.

DESCRIPTION

(11) References to items in the singular should be understood to include items in the plural, and vice versa, unless explicitly stated otherwise or clear from the text. Grammatical conjunctions are intended to express any and all disjunctive and conjunctive combinations of conjoined clauses, sentences, words, and the like, unless otherwise stated or clear from the context. Recitation of ranges of values herein are not intended to be limiting, referring instead individually to any and all

values falling within and/or including the range, unless otherwise indicated herein, and each separate value within such a range is incorporated into the specification as if it were individually recited herein. In the following description, it is understood that terms such as “first,” “second,” “top,” “bottom,” “side,” “front,” “back,” and the like are words of convenience and are not to be construed as limiting terms. For example, while in some examples a first side is located adjacent or near a second side, the terms “first side” and “second side” do not imply any specific order in which the sides are ordered.

(12) The terms “about,” “approximately,” “substantially,” or the like, when accompanying a numerical value, are to be construed as indicating a deviation as would be appreciated by one of ordinary skill in the art to operate satisfactorily for an intended purpose. Ranges of values and/or numeric values are provided herein as examples only, and do not constitute a limitation on the scope of the disclosure. The use of any and all examples, or exemplary language (“e.g.,” “such as,” or the like) provided herein, is intended merely to better illuminate the disclosed examples and does not pose a limitation on the scope of the disclosure. The terms “e.g.,” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. No language in the specification should be construed as indicating any unclaimed element as essential to the practice of the disclosed examples.

(13) The term “and/or” means any one or more of the items in the list joined by “and/or.” As an example, “x and/or y” means any element of the three-element set $\{(x), (y), (x, y)\}$. In other words, “x and/or y” means “one or both of x and y”. As another example, “x, y, and/or z” means any element of the seven-element set $\{(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)\}$. In other words, “x, y, and/or z” means “one or more of x, y, and z.”

(14) Disclosed is a low profile rotary damper assembly. In one example, a rotary damper assembly comprises a rotor, an annular seal, a housing, a bearing plate, and a gear. The rotor includes a shaft portion integrally formed with a body portion. The shaft portion defines an engagement interface at a distal end thereof. The annular seal is configured to surround at least a portion of the shaft portion. The housing defines an opening and a rotor cavity, wherein the housing is configured to rotatably support the rotor. The bearing plate is coupled to the housing and is configured to seal the opening. The bearing plate includes an annular shoulder defining an aperture and a convex cap defining a seal cavity for the annular seal between the body portion and an interior surface of the convex cap. The shaft portion extends axially outwardly through the aperture. The gear is coupled to the rotor via an opening that is configured to receive the engagement interface. The gear is shaped with a generally concave cavity that is sized and shaped to receive the collar and at least a portion of the convex cap.

(15) In some examples, each of the housing, the bearing plate, the rotor, and the gear are fabricated from a glass-reinforced polybutylene terephthalate (PBT).

(16) In some examples, the housing comprises a first cylindrical portion includes a first diameter and a second cylindrical portion includes a second diameter that is greater than the first diameter, wherein the first cylindrical portion houses the body portion of the rotor.

(17) In some examples, an outer surface of the first cylindrical portion comprises a plurality of teeth configured to engage a second gear.

(18) In some examples, the annular seal is compressed between and in contact with the annular shoulder and the body portion.

(19) In some examples, the rotary damper assembly further comprises a viscous material disposed within the rotor cavity. The viscous material can comprise a silicone oil or a silicone gel.

(20) In some examples, the housing includes a pivot shaft extending into the rotor cavity from a base of the first cylindrical portion. The pivot shaft can be configured to engage a cylindrical recess formed in the center cylindrical body such that the body portion can rotate about the pivot shaft.

(21) In some examples, the gear is fixedly mounted to the shaft portion at the engagement interface via a snap-in connection or an ultrasonic welding process. In some examples, at least one of the

gear or the shaft portion comprises an energy director.

(22) In some examples, the bearing plate is fixedly mounted to the housing via a snap-in connection or an ultrasonic welding process. The snap-in connection can comprise an annular recess and an annular rib for reception within the annular recess. The housing and/or bearing plate can comprise an annular energy director.

(23) In some examples, the engagement interface comprises a non-circular cross-section sized and shaped to correspond to a shape of the opening in the gear. The non-circular cross-section can have an x-shaped cross-section.

(24) In some examples, each of the housing, the bearing plate, the rotor, and the gear are fabricated from a glass-reinforced polybutylene terephthalate (PBT).

(25) FIG. 1a illustrates a topside isometric view of an example rotary damper assembly **100** in accordance with an aspect of this disclosure, while FIG. 1b illustrates an underside isometric view of the rotary damper assembly **100** of FIG. 1a. FIGS. 1c through 1f illustrate, respectively, first, second, third, and fourth elevation side views of the rotary damper assembly **100** of FIG. 1a. FIG. 1g illustrates a top plan view of the rotary damper assembly **100** of FIG. 1a, while FIG. 1h illustrates a bottom plan view thereof. FIG. 1i illustrates a cross-sectional elevation side view of the rotary damper assembly **100** taken along cutline A-A. FIG. 1j illustrates a cross-sectional isometric view of the rotary damper assembly **100** taken along cutline A-A. FIG. 1k illustrates an elevation side assembly view of the rotary damper assembly **100** of FIG. 1a, while FIG. 1l illustrates a cross-sectional view thereof taken along cutline A-A. FIG. 1m illustrates an isometric assembly view of the rotary damper assembly **100** of FIG. 1a, while FIG. 1n illustrates a cross-sectional view thereof taken along cutline A-A.

(26) As illustrated, the rotary damper assembly **100** generally comprises a housing **102**, a rotor **104**, a bearing plate **106**, an annular seal **108** (e.g., an O-ring, a toric joint, gasket, etc.), and a gear **110**. One or more components of the rotary damper assembly **100** (e.g., the gear **110**, the bearing plate **106**, the rotor **104**, and/or the housing **102**) can be fabricated from a plastic material using a plastic injection technique, additive manufacturing, or otherwise. For example, the rotary damper assembly **100**, or one or more components thereof, can be fabricated from glass-reinforced polybutylene terephthalate (PBT). In some examples, the glass-reinforced PBT is a 30% glass-reinforced PBT material, such as Celanex 3300-2. In some examples, one or more components of the rotary damper assembly **100** may be fabricated using material extrusion (e.g., fused deposition modeling (FDM), stereolithography (SLA), selective laser sintering (SLS), material jetting, binder jetting, powder bed fusion, directed energy deposition, VAT photopolymerisation, and/or any other suitable type of additive manufacturing/3D printing process.

(27) In operation, the rotary damper assembly **100** is configured to decelerate an objective movement of the gear **110** about an axis of rotation **112** relative to the housing **102**, which may be fixed or, as illustrated, geared. In the illustrated example, the rotor **104** includes a shaft portion **104a** and a body portion **104b**. In some examples, the shaft portion **104a** is integrally formed with the body portion **104b**. The body portion **104b** is configured to rotate within a viscous material housed within in the housing **102**, while the shaft portion **104a** extends outwardly from the housing **102**, through an aperture **114** in the bearing plate **106** (and through the annular seal **108**), to engage the gear **110** that is external to the housing **102**. The annular seal **108** is configured to surround at least a portion of the shaft portion **104a**. As illustrated, the annular seal **108** can be configured to surround a portion of the shaft portion **104a** at its proximal end (e.g., adjacent the body portion **104b**).

(28) In the illustrated example, the body portion **104b** comprises a center cylindrical body **116** surrounded by a plurality of outer walls **118** spaced from the center cylindrical body **116** to define an annular recess **120**. When the body portion **104b** rotates around the axis of rotation **112**, the viscous material inside the damper produces fluid resistance, which results in the damping torque. The damping torque can be affected by a number of factors. First, the damping torque generated is

a function of the size of the surface area of the rotor and housing disposed in contact with the viscous fluid. That is, the damping torque increases when the dimensions of the body portion **104b** is increased. Second, the viscosity of the type of viscous material is a variable for determining the amount of damper torque; however, the damper torque can also change as a function of the temperature and the rotation speed of the rotor **104**. For example, at higher temperatures, the torque decreases as the viscosity of the viscous material decreases at higher temperatures. Therefore, the viscous material can be any suitable viscous flowable fluid that may be desired for a particular application (e.g., based on operating temperature, and speed). In some examples, the viscous material is a silicone oil or a silicone gel. In some examples, the top surface of the rotor **104** includes one or more protrusions **154** (e.g., spacer bumps) to reduce wear between the rotor **104** and the bearing plate **106**.

(29) As best illustrated in the cross sectional views of FIGS. **1i**, **1j**, **1l**, and **1n**, the components of the rotary damper assembly **100** are configured to interconnect, while providing a low profile and without sacrificing the width of the gear teeth or reducing torque tolerance. In the illustrated example, the height (H) of the rotary damper assembly **100** is less than 20 mm, or about 14 mm.

(30) The illustrated rotary damper assembly **100** comprises a rotationally-symmetrical housing **102** defining a rotor cavity **122** in which a rotationally-symmetrical rotor **104** is mounted. As illustrated, the housing **102** defines an opening **124** and the rotor cavity **122**, where the rotor cavity **122** is configured to rotatably support the rotor **104**. The housing **102** includes a first cylindrical portion **102a** having a first diameter and a second cylindrical portion **102b** having a second diameter that is greater than the first diameter. The first cylindrical portion **102a** houses the body portion **104b** of the rotor **104**. The viscous material is disposed (and ultimately sealed) within the rotor cavity **122**.

(31) As illustrated, an outer surface (e.g., outer sidewall) of the first cylindrical portion **102a** comprises a plurality of teeth **126** configured to engage another gear (e.g., a second gear); however, other configurations are contemplated. For example, instead of a plurality of teeth, the second cylindrical portion **102b** can define a base with one or more flanges, tabs, and/or bolt-holes, which can be used to fixedly attach the housing **102** to a surface of a component.

(32) The housing **102** includes a pivot shaft **130** extending into the rotor cavity **122** of the housing **102** from the center of a base **128** of the first cylindrical portion **102a**. The pivot shaft **130** is configured to engage a cylindrical recess **132** formed in the center cylindrical body **116** such that the body portion **104b** can rotate about the pivot shaft **130** and about the axis of rotation **112**. Thus, the pivot shaft **130** can be concentric or otherwise align with the axis of rotation **112**. In the illustrated example, the pivot shaft **130** is configured to insert into a cylindrical recess **132** formed in the center cylindrical body **116** such that the body portion **104b** can rotate about the pivot shaft **130** and the axis of rotation **112**.

(33) The housing **102** further includes a plurality of curved walls **134** extending from the base **128** of the housing **102** towards the rotor cavity **122** of the housing **102**. The plurality of curved walls **134** is configured to rotate within the annular recess **120** formed in the body portion **104b** between a center cylindrical body **116** and outer walls **118**. The plurality of curved walls **134** and the plurality of outer walls **118** serve to increase the surface area of the rotor **104** and housing **102** that is in contact with the viscous fluid. In the illustrated example, four outer walls **118** and four curved walls **134** are illustrated, however additional or fewer outer walls **118** and/or curved walls **134** may be used depending on the desired damper torque.

(34) The bearing plate **106** is configured to couple to the housing **102** and configured to seal the opening **124**. For example, the second cylindrical portion **102b** can define an annular shoulder **136** with an inner annular lip **138** to engage an annular channel **150** of the bearing plate **106**. The bearing plate **106** comprises an annular recess configured to receive the inner annular lip **138** when supported by the annular shoulder **136**.

(35) The bearing plate **106** includes a collar **152** defining the aperture **114** and a convex cap **140** defining a seal cavity **142** for the annular seal **108**. The annular seal **108** is positioned between the

body portion **104b** and an interior surface of the convex cap **140**. When assembled, the annular seal **108** is compressed between and in contact with the collar **152** and the body portion **104b**. As illustrated, the shaft portion **104a** extends axially outwardly through the aperture **114**.

(36) The rotor **104** and housing **102** as well as the bearing plate **106** can be made of plastic material so that the components can be snapped (e.g., via a snap-in connection) or welded together (e.g., via ultrasonic welding). The snap-in connection can comprise an annular recess and an annular rib for reception within the annular recess. The annular recess can be on the housing **102** and the annular rib can be on the bearing plate **106** (or vice versa).

(37) In the illustrated example, however, the bearing plate **106** is fixedly mounted to the housing **102** via an ultrasonic welding process. To that end, an annular energy director **144** is formed on the shoulder-contacting surface of the bearing plate **106**. In another example, one or both of the housing **102** and the bearing plate **106** comprise an annular energy director **144**. The annular energy director **144** is used to facilitate the ultrasonic welding process by focusing and directing the ultrasonic energy to the desired welding location. As illustrated, the energy director is a small, raised feature and can be located on one or both of the bearing plate **106** and the housing **102**.

(38) The convex cap **140** of the bearing plate **106** is curved to define a seal cavity **142** for the annular seal **108** with a collar **152** at the center. The gear **110** is coupled to the rotor **104** via an opening **146** configured to receive the engagement interface **104c**. As illustrated, the gear **110** is shaped with a generally concave cavity **148** that is sized and shaped to receive the collar **152** and at least a portion of the convex cap **140**. By recessing at least a portion of the bearing plate **106**, the gear width (W) can be preserved, thus maintaining gear tooth strength. Further, as illustrated, a low profile is achieved, in part, because gear **110** is supported on underside of the gear **110** by convex profile of the convex cap **140**. Any deflection due to high torque and/or over gear mesh would be minimal by supporting the gear **110** with the convex cap **140**. As a result, the high torque has a minimal impact on the performance of the rotary damper assembly **100**.

(39) The illustrated rotor **104** has a shaft portion **104a** extending outwardly from the housing **102** and through the bearing plate **106**. In some examples, the shaft portion **104a** defines an engagement interface **104c** at a distal end thereof. As illustrated, the distal end of the shaft portion **104a** is shaped to define an engagement interface **104c** configured to engage a corresponding opening **146** in the gear **110**. The gear **110** is fixedly mounted to the shaft portion **104a** at the engagement interface **104c**. In some examples, the gear **110** is fixedly mounted to the shaft portion **104a** via an ultrasonic welding process and/or a snap-in connection.

(40) To provide an secure connection while also providing a small rotary damper assembly **100** (or to increase the connection strength, where desired), the engagement interface **104c** can use a non-circular cross-section and/or one of more engagement features (e.g., a complex profile and/or complex features, such as lobes) to increase the contact area between the gear **110** and the shaft portion **104a** at the engagement interface **104c**. A non-circular cross-section and/or one of more engagement features offers a number of advantages. First, a non-circular cross-section can increase the contact surface area between the between the gear **110** and the shaft portion **104a**, thus increasing the adhesive retention at the engagement interface **104c**. Second, a non-circular cross-section can be shaped to include engagement features, thus providing a higher degree of mechanical retention. For example, the gear **110** and/or the shaft portion **104a** may be shaped to form an engagement interface **104c** having one or more interlocking features to increase mechanical retention. Example interlocking features define a non-circular cross-section that is, as illustrated, a plus-shaped/x-shaped cross-section, but in other examples, it may comprise a non-circular cross-section with lobes, dovetails, etc. The corresponding opening **146** in the gear **110** is preferably shaped to correspond to the shape of the non-circular cross-section of the engagement interface **104c**. To that end, the engagement interface **104c** can comprise a non-circular cross-section sized and shaped to correspond to a shape of the opening **146** in the gear **110**.

(41) Once assembled, the gear **110** and the shaft portion **104a** can be joined via an ultrasonic

welding process. To that end, as best illustrated in FIGS. 2a and 2b, one or both of the gear 110 and the shaft portion 104a may have one or more energy directors 144 formed thereon (e.g., at the contact surfaces between the gear 110 and the shaft portion 104a). While a gear 110 is illustrated, other components and/or mechanical linkages are contemplated for otherwise connecting the rotary damper assembly 100 to or within a system having a component for which dampening control is to be provided.

(42) During assembly, the housing 102 is filled with a suitable viscous fluid, thereafter, the rotor 104 is inserted. Once the rotor 104 is inserted in the rotor cavity 122, the annular seal 108 is coupled to the shaft portion 104a. For example, the annular seal 108 can be slipped over and down the shaft portion 104a. The bearing plate 106 is coupled (e.g., via ultrasonic welding, snap-in connections, adhesives, or otherwise) to the housing 102 at the second cylindrical portion 102b to seal the viscous fluid within the rotor cavity 122. The gear 110 is then coupled (e.g., via ultrasonic welding, snap-in connections, adhesives, or otherwise) to the shaft portion 104a at its distal end (e.g., via the engagement interface 104c).

(43) While the present method and/or system has been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present method and/or system. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. For example, block and/or components of disclosed examples may be combined, divided, re-arranged, and/or otherwise modified. Therefore, the present method and/or system are not limited to the particular implementations disclosed. Instead, the present method and/or system will include all implementations falling within the scope of the appended claims, both literally and under the doctrine of equivalents.

Claims

1. A rotary damper assembly, comprising: a rotor having a shaft portion integrally formed with a body portion, wherein the shaft portion defines an engagement interface at a distal end thereof; an annular seal configured to surround at least a portion of the shaft portion; a housing defining an opening and a rotor cavity, wherein the housing is configured to rotatably support the rotor; a bearing plate coupled to the housing and configured to seal the opening, wherein the bearing plate includes a collar defining an aperture and a convex cap defining a seal cavity for the annular seal between the body portion and an interior surface of the convex cap, and wherein the shaft portion extends axially outwardly through the aperture; and a gear coupled to the rotor via an opening configured to receive the engagement interface, wherein the gear is shaped with a generally concave cavity that is configured to receive at least a portion of the convex cap and a cylindrical recess configured to receive the collar at least a portion of the convex cap.
2. The rotary damper assembly according to claim 1, wherein the housing comprises a first cylindrical portion having a first diameter and a second cylindrical portion having a second diameter that is greater than the first diameter, wherein the first cylindrical portion houses the body portion of the rotor.
3. A rotary damper assembly, comprising: a rotor having a shaft portion integrally formed with a body portion, wherein the shaft portion defines an engagement interface at a distal end thereof; an annular seal configured to surround at least a portion of the shaft portion; a housing defining an opening and a rotor cavity, wherein the housing is configured to rotatably support the rotor, and comprises a plurality of teeth on an outer surface thereof configured to engage a second gear; a bearing plate coupled to the housing and configured to seal the opening, wherein the bearing plate includes a collar defining an aperture and a convex cap defining a seal cavity for the annular seal between the body portion and an interior surface of the convex cap, and wherein the shaft portion

extends axially outwardly through the aperture; and a gear coupled to the rotor via an opening configured to receive the engagement interface, wherein the gear is shaped with a generally concave cavity that is sized and shaped to receive the collar and at least a portion of the convex cap.

4. The rotary damper assembly according to claim 1, wherein the annular seal is compressed between and in contact with an annular shoulder and the body portion.
 5. The rotary damper assembly according to claim 1, further comprising a viscous material disposed within the rotor cavity.
 6. The rotary damper assembly according to claim 5, wherein the viscous material comprises a silicone oil or a silicone gel.
 7. The rotary damper assembly according to claim 2, wherein the housing includes a pivot shaft extending into the rotor cavity from a base of the first cylindrical portion.
 8. The rotary damper assembly according to claim 7, wherein the pivot shaft is configured to engage a cylindrical recess formed in the body portion such that the body portion can rotate about the pivot shaft.
 9. The rotary damper assembly according to claim 1, wherein the gear is fixedly mounted to the shaft portion at the engagement interface via an ultrasonic welding process.
 10. The rotary damper assembly according to claim 8, wherein at least one of the gear or the shaft portion comprises an energy director.
 11. The rotary damper assembly according to claim 1, wherein the gear is fixedly mounted to the shaft portion at the engagement interface via a snap-in connection.
 12. The rotary damper assembly according to claim 1, wherein the bearing plate is fixedly mounted to the housing via an ultrasonic welding process.
 13. The rotary damper assembly according to claim 12, wherein the bearing plate comprises an annular energy director.
 14. The rotary damper assembly according to claim 12, wherein the housing comprises an annular energy director.
 15. The rotary damper assembly according to claim 1, wherein the bearing plate is fixedly mounted to the housing via a snap-in connection.
 16. The rotary damper assembly according to claim 15, wherein the snap-in connection comprises an annular recess and an annular rib for reception within the annular recess.
 17. The rotary damper assembly according to claim 1, wherein the engagement interface comprises a non-circular cross-section sized and shaped to correspond to a shape of the opening in the gear.
 18. The rotary damper assembly according to claim 17, wherein the non-circular cross-section has an x-shaped cross-section.
 19. The rotary damper assembly according to claim 1, wherein each of the housing, the bearing plate, the rotor, and the gear are fabricated from a glass-reinforced polybutylene terephthalate (PBT).
 20. A rotary damper assembly, comprising: a rotor having a shaft portion integrally formed with a body portion, wherein the shaft portion defines an engagement interface at a distal end thereof; an annular seal configured to surround at least a portion of the shaft portion; a housing configured to rotatably support the rotor, wherein the housing defines an opening, a rotor cavity, and a plurality of teeth formed on an exterior surface; a bearing plate coupled to the housing and configured to seal the opening, wherein the bearing plate includes a collar and a convex cap defining a seal cavity for the annular seal between the body portion and an interior surface of the convex cap, and wherein the shaft portion extends axially outwardly through an aperture in the bearing plate; and a gear coupled to the rotor via an opening configured to receive the engagement interface, wherein the gear is shaped with a generally concave cavity that is sized and shaped to receive the collar and at least a portion of the convex cap.
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